

EIP-1559 Compatibility & Txpool Analysis

Prepared by	JERUZZALEM – DAC Infra Tester
Date	Saturday, 09 May 2026
Environment	Dual Node — Windows (RPC :8545) + WSL/Linux (RPC :8546)
Report Type	Technical Findings — Transaction Compatibility Investigation
Relates to	Critical RPC Availability Issue (05 May 2026) · Testnet Infrastructure Validation Report (07 May 2026)

Executive Summary

This report documents a technical investigation into persistent transaction failures observed on DAC Testnet — specifically transactions that remained in queued, pending, or cancelled states across wallet clients. The investigation follows the infrastructure findings reported on 05 May 2026 (RPC availability) and 07 May 2026 (transaction propagation). This document extends those findings by identifying the root cause at the transaction format level and provides verifiable proof through node console output and RPC verification.

Finding	Description	Severity
EIP-1559 Incompatibility	MetaMask submits Type-2 tx; DAC validators do not reliably include them	Critical
eth_feeHistory Error	Node returns unmarshal error — MetaMask fee estimation degraded	High
Txpool Congestion	Network-wide backlog of 4,600–5,600 pending transactions	Medium
eth_call Stack Underflow	MetaMask contract probes fail — EVM compatibility gap	Low

Environment

Testing was conducted on a single-machine dual-node setup operating under CGNAT network constraints. Both nodes communicate internally via LAN routing, each connected outbound to DAC Testnet authority peers.

Node	Platform	RPC Endpoint	Coinbase
Windows Node	Windows Host	127.0.0.1:8545	0x2a01EAA...D3Af
WSL Node	WSL / Linux	127.0.0.1:8546	0x870ad6...AFE

Startup Commands

Windows Node:

```
.\dacnode.exe --identity "JERUZZALEM WINDOWS" --testnet --syncmode fast
--miner.etherbase 0x2a01EAA3175a2796acd8F64fCe6dD6488D39D3Af
--datadir "D:\AE - 903\APP\Miner\Airdrop\DAC\Windows\chaindata"
--port 28657 --maxpeers 12 --nat extip:192.168.100.7
--http --http.addr 127.0.0.1 --http.port 8545 --http.api eth,net,web3,txpool,personal
```

WSL Node:

```
./dacnode --identity "JERUZZALEM LINUX" --testnet --syncmode fast
--miner.etherbase 0x870ad63acc507cd8f878f170606d19ae78988AFE
--datadir ~/dac-chaindata-wsl --port 30304 --maxpeers 12
--nat extip:192.168.100.7 --http --http.addr 127.0.0.1
--http.port 8546 --http.api eth,net,web3,txpool,personal
```

Node Health Verification

Both nodes were attached via IPC and verified through the Geth JavaScript console prior to any transaction testing. Console output confirms node identity, sync status, nonce integrity, and txpool visibility.

Windows Node

```
instance: gdacnode/JERUZZALEM WINDOWS/v1.10.5-stable-e4023222/windows-amd64/go1.21.8
coinbase: 0x2a01eaa3175a2796acd8f64fce6dd6488d39d3af
at block: 14,793,815 (Fri May 08 2026 01:27:17 GMT+0700)

> eth.getTransactionCount("0x2a01...D3Af") → 59
> eth.getTransactionCount("0x2a01...D3Af", "pending") → 59
> txpool.status → { pending: 4649, queued: 798 }
```

WSL Node

```
instance: gdacnode/JERUZZALEM LINUX/v1.10.5-stable-e4023222/linux-amd64/go1.25.1
coinbase: 0x870ad63acc507cd8f878f170606d19ae78988afe
at block: 14,793,815 (Fri May 08 2026 01:27:17 GMT+0700)

> eth.getTransactionCount("0x870a...AFE") → 223
> eth.getTransactionCount("0x870a...AFE", "pending") → 223
> txpool.status → { pending: 4681, queued: 806 }
```

Both nodes show zero nonce discrepancy between confirmed and pending counts — confirming no stuck or ghost transactions in the local state. The txpool figures represent network-wide mempool visibility, not locally stuck transactions.

RPC Endpoint Verification

HTTP RPC endpoints were tested independently. PowerShell's native curl alias handles JSON escaping incorrectly and produces parse errors — Invoke-RestMethod was used on Windows instead.

Windows — Invoke-RestMethod

```
Invoke-RestMethod -Method POST -Uri http://127.0.0.1:8545
-ContentType "application/json"
-Body '{"jsonrpc":"2.0","method":"web3_clientVersion","params":[],"id":1}'

result → gdacnode/JERUZZALEM WINDOWS/v1.10.5-stable-e4023222/windows-amd64/go1.21.8
eth_blockNumber → 0xe1bc64 (decimal: 14,789,220)
```

WSL — curl

```
curl -X POST http://127.0.0.1:8546 -H "Content-Type: application/json"
-d '{"jsonrpc":"2.0","method":"web3_clientVersion","params":[],"id":1}'

result → gdnacnode/JERUZZALEM LINUX/v1.10.5-stable-e4023222/linux-amd64/go1.25.1
eth_blockNumber → 0xe1bc63 (decimal: 14,789,219)
```

Node	RPC Status	Block (hex)	Block (dec)
Windows :8545	Active	0xe1bc64	14,789,220
WSL :8546	Active	0xe1bc63	14,789,219

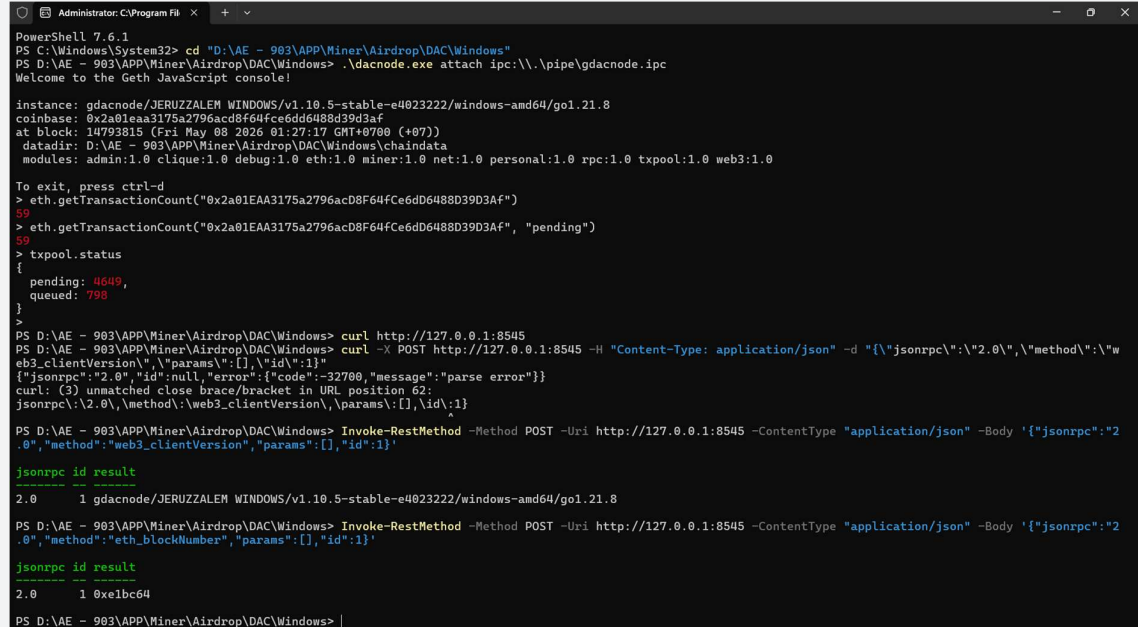
The 1-block difference between nodes is within normal block propagation latency. Both nodes are on the same canonical chain and actively advancing.

Redemonstrate — Supporting Screenshots

The following screenshots were captured during a post-investigation redemonstration to validate that the dual-node setup remained stable and the findings were reproducible.

Fig. 1 — Windows Node: IPC Console, Txpool Status & RPC Verification

Confirms: identity, coinbase, nonce 59 (confirmed = pending, no stuck tx), txpool pending 4,649 / queued 798, Invoke-RestMethod returning web3_clientVersion and eth_blockNumber (0xe1bc64).



```
Administrator: C:\Program Files\PowerShell 7.6.1
PS C:\Windows\System32> cd "D:\AE - 983\APP\Miner\Airdrop\DAC\Windows"
PS D:\AE - 983\APP\Miner\Airdrop\DAC\Windows> .\gdnacnode.exe attach ipc:\\.\pipe\gdnacnode.ipc
Welcome to the Geth JavaScript console!

instance: gdnacnode/JERUZZALEM WINDOWS/v1.10.5-stable-e4023222/windows-amd64/go1.21.8
coinbase: 0x2a01eaa3175a2796acd8f64fce6dd6488d39d3af
at block: 14793815 (Fri May 08 2026 01:27:17 GMT+0700 (+07))
datadir: D:\AE - 983\APP\Miner\Airdrop\DAC\Windows\chaindata
modules: admin:1.0 clique:1.0 debug:1.0 eth:1.0 miner:1.0 net:1.0 personal:1.0 rpc:1.0 txpool:1.0 web3:1.0

To exit, press ctrl-d
> eth.getTransactionCount("0x2a01EAA3175a2796acd8f64fCe6dD6488D39D3Af")
59
> eth.getTransactionCount("0x2a01EAA3175a2796acd8f64fCe6dD6488D39D3Af", "pending")
59
> txpool.status
{
  pending: 4649,
  queued: 798
}
>
PS D:\AE - 983\APP\Miner\Airdrop\DAC\Windows> curl http://127.0.0.1:8545
PS D:\AE - 983\APP\Miner\Airdrop\DAC\Windows> curl -X POST http://127.0.0.1:8545 -H "Content-Type: application/json" -d '{"jsonrpc":"2.0","method":"web3_clientVersion","params":[],"id":1}'
{"jsonrpc":"2.0","id":null,"error":{"code":-32700,"message":"parse error"}}
curl: (3) unmatched close brace/bracket in URL position 62:
jsonrpc:\2.0\,method:\web3_clientVersion\,params:[],id:1}

PS D:\AE - 983\APP\Miner\Airdrop\DAC\Windows> Invoke-RestMethod -Method POST -Uri http://127.0.0.1:8545 -ContentType "application/json" -Body '{"jsonrpc":"2.0","method":"web3_clientVersion","params":[],"id":1}'

jsonrpc id result
2.0 1 gdnacnode/JERUZZALEM WINDOWS/v1.10.5-stable-e4023222/windows-amd64/go1.21.8

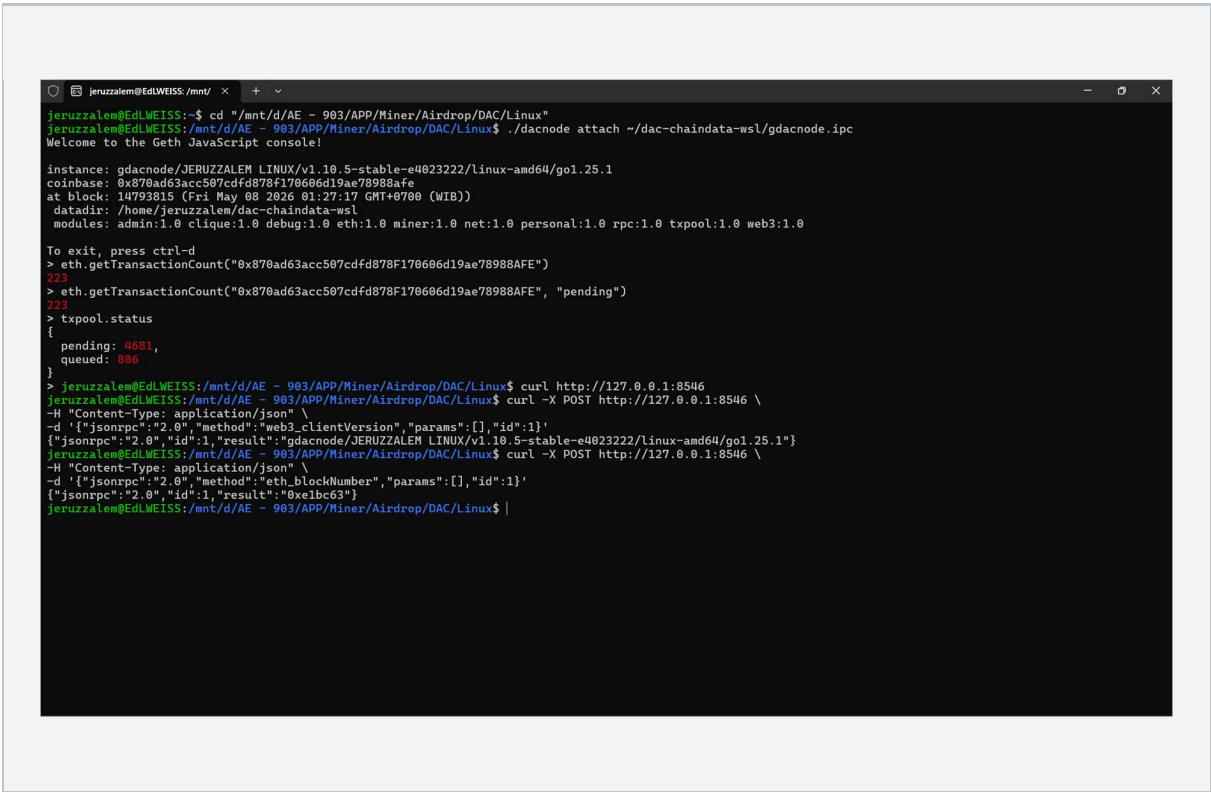
PS D:\AE - 983\APP\Miner\Airdrop\DAC\Windows> Invoke-RestMethod -Method POST -Uri http://127.0.0.1:8545 -ContentType "application/json" -Body '{"jsonrpc":"2.0","method":"eth_blockNumber","params":[],"id":1}'

jsonrpc id result
2.0 1 0xe1bc64

PS D:\AE - 983\APP\Miner\Airdrop\DAC\Windows>
```

Fig. 2 — WSL Node: IPC Console, Txpool Status & RPC Verification

Confirms: identity, coinbase, nonce 223 (confirmed = pending, no stuck tx), txpool pending 4,681 / queued 806, curl returning web3_clientVersion and eth_blockNumber (0xe1bc63).



Transaction Compatibility Test

Cross-wallet transfers were performed in both directions between the two node-wallet pairs to validate end-to-end transaction flow through the local RPC infrastructure.

Direction	From	To	Wallet	Result
A → B	0x2a01...D3Af	0x870a...AFE	OKX Wallet	Confirmed
B → A	0x870a...AFE	0x2a01...D3Af	MetaMask	Pending

Node Logs at Time of Submission

Windows Node:

```
WARN Served eth_feeHistory err="invalid argument 0: json: cannot unmarshal string into Go value of type int"
INFO Imported new chain segment blocks=1 txs=117 mgas=7.934 elapsed=83.259ms
INFO Submitted transaction hash=0xb57f...f7214
from=0x2a01...D3Af nonce=38 recipient=0x870a...AFE
```

WSL Node:

```
WARN Served eth_call err="stack underflow (0 <=> 7)"
INFO Submitted transaction hash=0xf629...b0ac
    from=0x870a...AFE nonce=196 recipient=0x2a01...D3Af
```

Root Cause Analysis

EIP-1559 Type-2 Transaction Incompatibility

Inspection of the pending MetaMask transaction via `eth.getTransaction()` revealed the root cause:

```
eth.getTransaction("0xf629cb2a81a0d361cfcd53d003749f14b34dc4a136a065e72831b9389a61b0ac")

{
  type:           "0x2",
  gasPrice:       0,
  maxFeePerGas:   4015726755676,
  maxPriorityFeePerGas: 4015726755676,
  nonce:          196,
  gas:            21000,
  chainId:        "0x5586",
  blockHash:      null,
  blockNumber:    null
}
```

The transaction is type 0x2 — an EIP-1559 transaction. `gasPrice` is 0 and fee parameters are expressed via `maxFeePerGas` and `maxPriorityFeePerGas`. `blockHash` and `blockNumber` are null, confirming the transaction was never included in a block despite being in the mempool.

DAC Testnet is based on Geth v1.10.5. The repeated `eth_feeHistory` errors indicate incomplete EIP-1559 support. MetaMask defaults to EIP-1559 transactions and relies on `eth_feeHistory` for fee estimation — when this call fails, the resulting transaction parameters are not accepted by DAC validators. OKX Wallet submits legacy Type-0 transactions which are fully compatible with the current validator set.

Additional Indicators

Observation	Interpretation
<code>eth_feeHistory</code> → unmarshal error on every MetaMask call	Node expects int, receives string — partial EIP-1559 implementation
<code>eth_call</code> → stack underflow (0 <=> 7)	MetaMask contract probes fail — EVM gap; does not block ETH transfers directly
OKX confirmed / MetaMask pending under identical conditions	Transaction type is the sole differentiating variable
<code>tx.blockHash</code> = null after extended time in mempool	Validators are not selecting type-2 transactions for inclusion

Txpool Congestion

At the time of investigation, both nodes reported significantly elevated network-wide txpool state. A pending count in the thousands on a testnet is abnormally high and constitutes a secondary contributing factor to transaction reliability issues — even valid, compatible transactions experience extended confirmation delays.

Node	Pending	Queued
Windows :8545	4,649	798
WSL :8546	4,681	806

Infrastructure Status

Component	Status	Notes
Windows node (v1.10.5)	Healthy	Synced, peered, HTTP RPC active
WSL node (v1.10.5)	Healthy	Synced, peered, HTTP RPC active
Local RPC :8545	Active	Confirmed via Invoke-RestMethod
Local RPC :8546	Active	Confirmed via curl
Block sync	Healthy	1-block delta — normal propagation
OKX Wallet transactions	Confirmed	Legacy tx — validator compatible
MetaMask transactions	Pending	EIP-1559 type-2 — not included by validators
DAC network txpool	Congested	4,600–5,600 pending network-wide
eth_feeHistory	Error	Unmarshal type mismatch in node handler
eth_call (MM probes)	Error	Stack underflow — EVM compatibility gap

Conclusions

- Local node infrastructure is not the cause of transaction failures.**

Both nodes are healthy, synced, and capable of broadcasting and receiving transactions. The local RPC layer functions correctly on both Windows and WSL environments.
- The root cause of MetaMask transaction failures is EIP-1559 incompatibility.**

DAC Testnet validators do not reliably include Type-2 (EIP-1559) transactions. The incomplete eth_feeHistory implementation compounds this by degrading MetaMask's fee estimation.
- OKX Wallet and Rabby Wallet are significantly more compatible.**

These wallets submit legacy-style transactions that the current DAC validator set handles correctly. Switching wallet is the most immediate remediation available to users.

4 **Network congestion is a secondary contributing factor.**

Even compatible transactions experience delays due to a network-wide pending backlog, suggesting validator throughput limitations independent of transaction type.

Recommendations

For Testnet Participants

- Use OKX Wallet or Rabby Wallet as primary wallets — avoid MetaMask until EIP-1559 compatibility is confirmed.
- Run a local node and connect wallets to a self-hosted RPC endpoint for improved reliability.

For DAC Team

- Audit and complete eth_feeHistory implementation — fix the unmarshal type mismatch.
- Validate Type-2 (EIP-1559) transaction handling at the validator and txpool propagation layer.
- Publish wallet compatibility guidance for testnet participants.
- Investigate txpool congestion — evaluate validator throughput and block gas limits.
- Implement monitoring for RPC latency, timeout rate, and txpool queue depth.